

Multi-Scale Causal Convolution-Based Swin Transformer for Motor Imagery EEG Classification

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Abstract—Motor Imagery (MI) classification, a core technique of brain-computer interface (BCI) systems, aims to accurately recognize electroencephalogram (EEG) patterns generated when users imagine specific motor actions. However, traditional approaches face limitations in effectively capturing spatiotemporal features and modeling long-range dependencies in EEG signals. To address these challenges, this paper proposes a hybrid architecture that integrates convolutional neural networks (CNNs) and the Swin Transformer, termed the Causal Convolutional Swin Transformer Network (CSTNet). CSTNet leverages the local-feature extraction strength of CNNs and the Swin Transformer's capacity to capture global contextual dependencies. Specifically, CSTNet employs a multi-scale causal convolution module to extract time-frequency features from EEG signals, then uses depthwise separable convolution to capture spatial patterns. Subsequently, the hierarchical window-based self-attention of the Swin Transformer models long-range temporal dependencies, and a global self-attention layer further enhances feature representations. The network processes EEG signals end-to-end. In evaluation on the publicly available BCI Competition IV-2a dataset, experimental results demonstrated that CSTNet achieved an average classification accuracy of 83.45% and outperformed several state-of-the-art baseline networks. Taken together, these results validated the effectiveness of the proposed approach for MI classification and provided a promising technical pathway for practical BCI applications.

Index Terms—Multi-scale Causal Convolution; Swin Transformer; Global Self-attention; EEG; BCI

I. INTRODUCTION

With the rapid development of deep learning technologies, significant breakthroughs have been achieved in bioelectrical signal processing, particularly in electroencephalography (EEG) analysis. Motor imagery (MI), the cognitive process of activating the sensorimotor cortex by imagining limb movements without actual execution, has become a core technique in brain-computer interface (BCI) research. The goal of MI-based BCI systems is to accurately recognize EEG patterns and thereby enable intention-driven control of external devices, with broad applications in neural rehabilitation, assistive communication, and intelligent support systems. However, due to the low signal-to-noise ratio, strong non-stationarity, spatial redundancy, and substantial inter-subject variability of EEG signals, extracting efficient and discriminative features remains a major challenge.

Despite these challenges, the end-to-end automatic feature-learning paradigm of deep learning has opened new prospects for EEG decoding; nevertheless, it still exhibits certain limitations. SaleNet [1] is a low-power end-to-end CNN model developed for continuous attention-level assessment from EEG. EEG-Inception [2] is an end-to-end neural network that achieves high accuracy and robustness and was specifically designed for EEG-based motor imagery classification. Models such as ShallowCNN and DeepCNN [3], along with the compact and efficient EEGNet [4], have also produced promising results in EEG classification tasks. However, convolutional neural networks are primarily effective at capturing local spatiotemporal patterns and are insufficient for modeling long-term temporal dependencies, while Transformer-based architectures [5], although powerful for global modeling, often incur high computational costs and may overlook fine-grained details.

Building on these observations, this study proposes a novel end-to-end hybrid architecture, the Causal Multi-Scale Swin Transformer Network (CSTNet). The architecture introduces a multi-scale causal convolution module that integrates three parallel branches of CausalConv [6], thereby preserving temporal causality while focusing on multi-scale local features of EEG signals. Spatial compression and patch segmentation are further performed using depthwise group convolution and dual average pooling, providing an appropriate granularity for subsequent sequence modeling. The model's backend adopts a hierarchical Swin Transformer structure [7], whose shifted-window attention mechanism balances local and global information interaction and efficiently captures long-range temporal dependencies. In addition, a global self-attention layer is incorporated to further improve global feature modeling.

The local attention mechanism in CSTNet, implemented through window partitioning and shifting operations, effectively balances detailed local representation with global contextual information. In particular, local attention emphasizes dynamic variations over short time spans and enhances the model's ability to detect weak and transient EEG features. Meanwhile, the shifted-window mechanism facilitates information flow across windows, thereby improving the modeling of long-range dependencies. The global self-attention layer compensates for local attention's limitations in global mod-

eling and enables flexible integration of information across multiple scales, improving overall discriminative performance.

To validate the proposed approach, we evaluated CSTNet on the BCI Competition IV-2a dataset [8] under subject-dependent MI classification settings. The experimental results demonstrated that CSTNet consistently outperformed established baselines such as ShallowCNN, EEGNet, and EEG Conformer. Two primary contributions of this study are summarized as follows.

- We design a multi-scale causal convolutional front-end to effectively extract EEG features across different temporal scales while preventing information leakage from future time points, thereby enhancing the model's capacity to represent temporal dependencies.
- By incorporating shifted-window attention and global self-attention mechanisms, the proposed framework achieves a balanced trade-off between computational efficiency and modeling capacity, which in turn significantly improves the accuracy and generalizability of motor imagery EEG decoding.

II. RELATED WORK

A. Multi-scale Causal Convolutional Network

In EEG decoding research, convolutional neural networks (CNNs) are widely adopted in motor imagery (MI)-based BCI systems because of their strong capability for local feature extraction. Early architectures such as ShallowCNN and DeepCNN effectively capture spatiotemporal patterns in EEG signals by stacking temporal and spatial convolutional layers. EEGNet further introduces depthwise separable convolution, which substantially reduces the number of parameters while improving generalization performance. LMDA-Net's deep attention module employs semi-global pooling and local cross-depth interactions to dynamically reweight features across multiple dimensions, thereby enabling efficient feature fusion [9]. However, conventional CNNs typically rely on single-scale kernels and therefore struggle to capture discriminative information that occurs at different temporal scales.

To address this limitation, multi-scale convolution methods such as MSCNN [10] have been proposed; by running convolution kernels of multiple sizes in parallel within the same layer, these methods can capture short-term transient events as well as slow-varying long-term trends. The effectiveness of multi-scale modeling has also been validated by applications such as MS-TCN in sleep staging [11]. Moreover, MSCFormer employs multi-branch CNNs to extract spatiotemporal features at different scales and then leverages a Transformer encoder to integrate global dependencies, thereby mitigating inter-subject variability in EEG signals [12].

Furthermore, novel architectures based on causal convolution have also been developed to address multi-subject motion modeling and generation tasks; these architectures capture intrinsic motion characteristics across diverse subjects [13]. Causal convolutional architectures enforce temporal causality through one-sided (causal) padding to prevent information

leakage from future time steps, making them well suited to sequential tasks such as speech recognition and electrocardiogram (ECG) analysis. Combining multi-scale convolution with causal convolution improves a model's sensitivity to cross-scale patterns while preserving real-time operation and physiological plausibility.

B. Swin-Global Network

Turning to attention-based models, Transformer architectures have attracted considerable attention in the EEG domain due to their self-attention mechanisms and strong global modeling capabilities. EEG-Transformer applies multi-head self-attention to MI decoding and significantly enhances the modeling of long-range dependencies [14]. However, the computational complexity of a standard Transformer grows quadratically with sequence length, making it difficult to handle high-sampling-rate, long-duration EEG signals. To address this challenge, approaches such as ViT and PatchTST divide sequences into patches and share attention weights, thereby improving computational efficiency [15]. EEG Conformer extracts local features via one-dimensional spatiotemporal convolutional layers and then employs self-attention modules to capture global associations among those local features [16]. The shifted-window attention mechanism introduced in the Swin Transformer computes self-attention within local windows and enables inter-window communication, effectively reducing computational complexity so that it scales linearly with sequence length. Consequently, lightweight Transformer variants based on localization and adaptive window strategies have emerged as practical solutions for modeling long-range dependencies in EEG.

Nevertheless, although global attention enhances a model's ability to capture long-range dependencies, relying solely on global modeling can neglect local details. Consequently, recent research focuses on local-global fusion mechanisms. For example, CTNet uses convolutional modules to extract spatial features and then applies a Transformer encoder's multi-head attention to capture global dependencies among higher-level representations [17]. TCANet combines temporal convolutional layers to extract dynamic temporal features, employs attention mechanisms to amplify the importance of critical time steps, and introduces enhanced residual connections to improve feature propagation from shallow to deep layers [18].

III. METHODS

To enhance the representational capacity for spatiotemporal features of motor imagery electroencephalography (MI-EEG) and the modeling capability for long-range dependencies, the present study develops a novel end-to-end neural network architecture that combines multi-scale causal convolution and hierarchical Transformer structures—namely, the Causal Multi-Scale Swin Transformer Network (CSTNet). The network comprises three key components: a Multi-Scale Causal Convolution embedding module, a Swin Transformer module for local modeling, and a Global Multi-Head Self-Attention

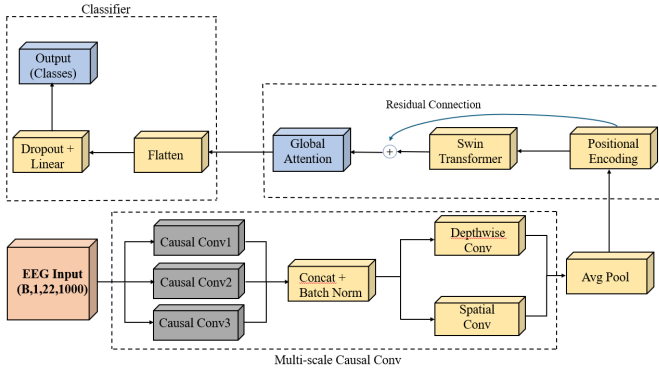


Fig. 1. Proposed causal multi-scale swin transformer network architecture

(MHSA) module for global modeling. Figure 1 illustrates the detailed architecture.

A. Multi-Scale Causal Convolution

To extract fine-grained dynamic variations of EEG signals at different temporal scales, the front end of CSTNet is designed with three parallel causal convolution branches, each employing convolutional kernels of different sizes to capture temporal features at large, medium, and small receptive fields. This design thereby guarantees that the output of each convolutional branch depends only on current and past samples, satisfying temporal causality for EEG analysis and avoiding leakage of future information. The branch outputs are computed in the following way:

$$F_i = \text{CausalConv}(x; k_i), \quad i = 1, 2, 3, \quad (1)$$

where $\mathbf{x} \in \mathbb{R}^{B \times 1 \times C \times T}$ denotes the raw EEG input, k_i denotes the kernel size of the i -th branch, C is the number of channels, T is the number of time steps, and B is the batch size. The outputs of the three-scale convolutions are concatenated along the channel dimension and are then sequentially passed through batch normalization, a depthwise separable convolution, and a spatial convolution module to achieve spatial compression and feature integration. Finally, average pooling and dropout are applied to reduce redundancy and obtain the temporal embedding used for subsequent Transformer encoding. CSTNet is founded on this principle: its front end uses three parallel causal-convolution branches with different receptive fields to balance multi-scale dynamic features and temporal causality, thereby laying a solid foundation for subsequent global modeling.

B. Swin Transformer Network

After positional encoding, the embedded features are fed into a hierarchical Swin Transformer module. The module employs the shifted-window attention mechanism to maintain efficient local modeling while enabling information exchange across windows, thereby mitigating the information isolation

caused by conventional windowed attention. The windowed self-attention operation is computed in the following way:

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^\top}{\sqrt{d_k}} + B_{\text{rel}}\right)V, \quad (2)$$

where B_{rel} denotes the relative position bias used to enhance positional sensitivity, and the query, key and value projections are obtained via linear projection matrices Q, K, V . To enable cross-window modeling, window shifting is applied alternately every two Transformer blocks, so that information propagates across different local receptive fields.

Residual connections and layer normalization are applied within each block in the following way:

$$x' = \text{LayerNorm}(x + \text{WindowAttention}(x)), \quad (3)$$

$$x'' = \text{LayerNorm}(x' + \text{MLP}(x')). \quad (4)$$

Stacking multiple such blocks yields hierarchical local modeling capability; as a result, the model can extract multi-granularity semantic features with relatively low computational cost.

C. Global Self-Attention Structuring

Although the Swin Transformer can model cross-region dependencies to some extent, its reliance on local window partitioning limits efficient global feature integration. To complement this limitation, CSTNet further introduces a Global Multi-Head Self-Attention module after the Swin stages. This module performs unconstrained global dependency modeling on the full feature sequence and can directly establish relationships between any two positions in the sequence. Concretely, the global self-attention module operates on the complete feature sequence $\mathbf{F}_{\text{swin}} \in \mathbb{R}^{B \times L \times D}$ in the following way:

$$F_{\text{global}} = \text{LayerNorm}(F_{\text{swin}} + \text{MHSA}(F_{\text{swin}})), \quad (5)$$

where B denotes the batch size, L the sequence length, and D the feature dimensionality. Specifically, the module splits the feature space into multiple subspaces via multiple heads; each head independently computes queries, keys and values and then produces scaled dot-product attention weights, followed by residual connections and normalization. The global attention mechanism effectively enhances the model's ability to represent long-range dependencies and global context, which is particularly beneficial for integrating information across channels and time segments in complex motor imagery tasks.

CSTNet incorporates a hierarchical Swin Transformer in the back end and adds a global self-attention layer to further strengthen global feature integration. The inclusion of local attention mechanisms allows the model to flexibly fuse information across multiple scales, balancing local dynamics and global dependencies and thereby providing a more efficient and accurate modeling scheme for MI-EEG decoding.

D. Classification Network

CSTNet's novelty lies in its multi-level feature fusion strategy, which integrates complementary representations extracted at different stages of the network. Specifically, the model fuses three key feature streams according to the following rationale: the positional embedding F_{pos} preserves fine-grained time-frequency and spatial-topological information of the original signal; the Swin Transformer output F_{swin} captures multi-scale local patterns and cross-window mid-range dependencies; and the global attention output F_{global} provides long-range dependencies and full-sequence context. These three components are fused residually in the following way:

$$F_{\text{final}} = F_{\text{pos}} + F_{\text{swin}} + F_{\text{global}}. \quad (6)$$

The classification head comprises two stages. First, a dropout layer is applied for regularization to mitigate overfitting. Second, a fully connected linear layer maps the flattened high-dimensional representation to the class logits; a softmax operator then yields the final class probabilities:

$$\hat{y} = \text{Softmax}(\text{FC}(\text{Flatten}(F_{\text{final}}))). \quad (7)$$

CSTNet's three-tier progressive architecture—temporal aware multi-scale causal convolutional feature extraction, spatial and channel modeling via Swin Transformer, and global integration via attention enhancement—reflects a systematic approach to local-global-cross-scale modeling for EEG signals. Beginning from fine-grained time-frequency features, the hierarchical strategy progressively constructs mid-range spatiotemporal patterns and ultimately integrates global long range dependencies. This results in a comprehensive multi scale feature representation that substantially improves the accuracy and generalizability of motor imagery classification and offers a more expressive neural modeling paradigm for BCI tasks.

IV. EXPERIMENTS

A. BCI IV-2a Dataset

BCI IV-2a dataset consists of EEG recordings from nine subjects using 22 EEG channels. The data were collected at a sampling rate of 250 Hz across two sessions, each comprising 288 trials.

B. Analysis of experimental results under the BCI IV-2a dataset

To systematically evaluate the classification performance of the CSTNet model proposed in this study on the BCI Competition IV-2a dataset, we compared CSTNet's performance against several state-of-the-art deep learning models for this classification task, including LMDA, ShallowCNN, DeepCNN, EEGNet, EEG Conformer, MSCFormer, CTNet, and TCANet. Among these, we reimplemented LMDA, ShallowCNN, DeepCNN, EEGNet, CTNet, and EEG Conformer under a unified data-preprocessing and training protocol, whereas the reported performance metrics for the remaining

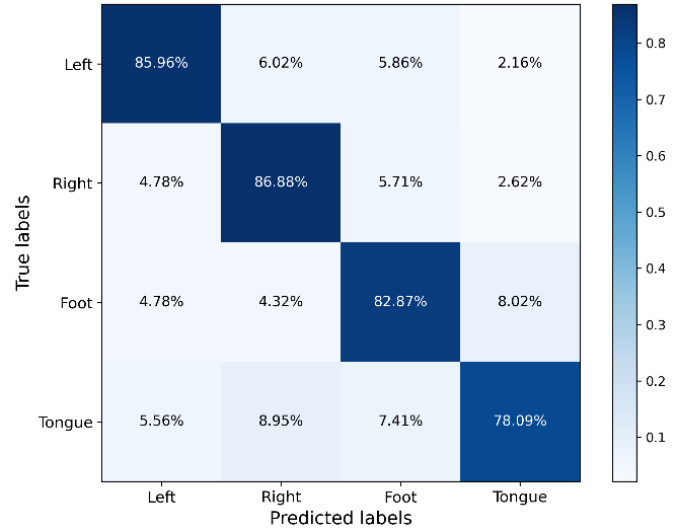


Fig. 2. Confusion matrix for subject-specific experimental results.

models were taken from their original publications to ensure fairness and rigor in cross-model comparisons.

As shown in Table 1, CSTNet achieved the best overall performance among all models, with a mean classification accuracy of 83.45% ($\pm 8.98\%$), and obtained the highest accuracies for subjects S03, S07, and S08. CSTNet also achieved the highest Kappa coefficient (0.7793), outperforming traditional convolutional networks such as ShallowCNN (75.27%, $p = 0.0010$) and DeepCNN (77.43%, $p = 0.0298$), and showing statistically significant improvements over attention-based models such as EEGNet (77.55%, $p = 0.0141$) and EEG Conformer (77.81%, $p = 0.0237$). Compared with CTNet (81.37%, $p = 0.0099$) and LMDA (67.05%, $p = 0.0001$), CSTNet likewise demonstrated a statistically significant advantage. Relative to MSCFormer (82.95%) and TCANet (83.06%), CSTNet's mean accuracy increased by 0.50% and 0.39%, respectively. Moreover, CSTNet achieved the best results on multiple subjects and showed the greatest consistency in performance across models, further confirming its superior generalization capability.

The single-subject experimental results on the BCI Competition IV-2a dataset were presented in Figure 2. The confusion matrices showed high diagonal values for all four classes, indicating strong class discrimination. Specifically, the recall rates for the left-hand and right-hand classes were comparable and both exceeded 85%, demonstrating stable performance. CSTNet integrates the dual advantages of spatial convolution and temporal modeling: by adopting a hierarchical architecture, it performs multi-scale feature extraction and temporal relationship modeling for EEG signals, thereby enhancing representational capacity without substantially increasing computational cost. This strategy enables the model to more effectively capture discriminative patterns in motor imagery signals, resulting in superior accuracy and stability compared with existing mainstream methods.

TABLE I
COMPARISON OF RESULTS OF SUBJECT-SPECIFIC CLASSIFICATION EXPERIMENTS

Models	S01	S02	S03	S04	S05	S06	S07	S08	S09	Avg±Std	Kappa	P-Val
LMDA [9]	77.18	47.22	90.97	60.07	66.67	53.82	71.18	75.69	66.67	67.05±12.01	0.6145	0.0001
ShallowCNN [3]	84.72	57.64	92.36	70.83	69.79	61.11	76.39	84.72	79.86	75.27±10.82	0.6713	0.0010
DeepCNN [3]	80.56	48.26	88.54	77.08	73.96	68.40	90.63	84.72	84.72	77.43±12.28	0.6132	0.0298
EEGNet [4]	86.81	63.54	91.67	69.79	73.26	52.78	88.19	84.72	87.15	77.55±12.61	0.7006	0.0141
EEG Conformer [16]	85.26	50.36	91.32	77.37	76.08	64.25	88.65	86.15	80.86	77.81±13.11	0.7135	0.0237
MSCFormer [12]	86.11	65.42	94.10	85.97	80.42	74.58	89.93	84.79	85.21	82.95±8.06	0.7726	0.6863
TCANet [18]	88.89	70.14	92.71	79.86	77.78	74.31	92.71	85.76	85.42	83.06±8.02	0.7742	0.6494
CTNet [17]	87.16	69.79	92.36	84.03	74.30	63.89	89.58	85.07	86.11	81.37±9.71	0.7515	0.0099
CSTNet	87.85	71.53	95.14	82.99	77.08	69.44	92.36	87.85	86.81	83.45±8.98	0.7793	—

While ablation studies could have offered a more detailed understanding of the contributions of individual modules, the significant computational cost and time requirements placed such experiments beyond the current scope. Instead, we provided theoretical analysis and baseline (CTNet) comparisons to support the effectiveness of the proposed design, and we left detailed ablation analysis for future research.

V. CONCLUSIONS

This study proposes a hybrid network, the Causal Multi-Scale Swin Transformer Network (CSTNet), for motor imagery classification based on electroencephalography. The network employs a multi-scale causal convolution module to precisely extract the time–frequency features embedded in EEG signals, leverages depthwise separable convolutions to efficiently capture spatial distribution patterns, and utilizes the Swin Transformer’s hierarchical shifted-window attention mechanism to explicitly model long-range temporal dependencies. A global self-attention layer is additionally incorporated to further enhance the representation of critical features, ultimately enabling accurate classification of motor imagery categories. Multi-scale causal convolution, depthwise separable convolution, and the Swin Transformer’s hierarchical attention mechanism are deeply fused and applied to the domain of EEG-based motor imagery classification. The hybrid network was validated on the standard BCI Competition IV-2a dataset, and the results showed that it achieved superior classification accuracy in subject-specific experiments. Although this work adopted a topic-dependent classification setting without cross-topic testing, we acknowledge that considerable variability exists across different topics. The choice of topic-dependent evaluation was primarily motivated by its practicality and by its consistency with previous studies that emphasize within-topic reliability. Nevertheless, future work should investigate cross-topic generalization to provide a more comprehensive evaluation of model robustness.

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